UKUDLA Data Science Certificate

Data Visualization

Markus Mößler

University of Hohenheim

UKUDLA
African German Centre
for Sustainable and Resilient Food Systems and Applied Agricultural and Food Data Science

Supported by:

Federal Foreign Office

With funding from the:

Federal Ministry of Research, Technology and Space

With support from

Federal Ministry of Agriculture, Food and Regional Identity

by decision of the German Bundestag

science, technology & innovation
Department: Science, Technology and Innovation
REPUBLIC OF SOUTH AFRICA

NRF National Research Foundation

DAAD Deutscher Akademischer Austauschdienst German Academic Exchange Service

Table of contents

UKUDLA

Prepared data do not reveal their patterns by themselves

A table preserves exact values

- countries and years are explicit;
- units and variables are documented;
- observations can be queried; and
- computations are reproducible.

A visual representation can reveal

- distributions and unusual values;
- differences between countries;
- change over time; and
- relationships between variables.

Data visualization maps documented observations to visual properties so that patterns can be inspected and communicated.

A chart is an analytical argument

Question and audience → Selection, encoding, scale and grouping → Visible pattern and comparison → Interpretation and limitations

- Exploratory graphics support investigation and iteration.
- Explanatory graphics focus attention on a defensible message.
- Both depend on choices about inclusion, aggregation and visual encoding.
- Neither turns an association into a causal effect.

This session moves from **motivation**, through essential **concepts**, to a reproducible **application** in an analytical workflow.

Start with the analytical question

Question	Relevant structure	Possible graphic
How are values distributed?	One quantitative variable	Histogram, density plot, boxplot
How do groups compare?	Quantitative value and category	Aligned points, boxplots, faceting
How does a measure change?	Quantitative value ordered by time	Line or point plot
How do two measures vary together?	Two quantitative variables	Scatterplot
Where does a measure occur?	Measure and valid spatial geometry	Map

Choose a graphic from the question and variable types—not from a menu of decorative chart forms.

Build a graphic from explicit components

Data and observation grain → Aesthetic mappings → Geometric marks → Scales and coordinates → Facets, labels and guides

- Map variables to position, colour, shape, size or grouping.
- Select marks—points, lines, bars or areas—that support comparison.
- Use faceting when small multiples clarify group differences.
- Add labels and annotations that make definitions and units visible.

In `ggplot2`, every visible element should follow from an explicit layer or mapping.

UKUDLA

Use the strongest encodings for important comparisons

Usually easier to compare

1. position on a common scale;
2. position on aligned scales;
3. length; and
4. angle or area.

Use with care

- colour hue distinguishes categories;
- colour lightness can show order;
- size is read through area, not radius;
- shape supports few categories; and
- redundant cues improve accessibility.

Encode the quantity that matters most with position when possible; reserve colour and size for distinctions they can carry reliably.

Scales and aggregation change what the viewer sees

Choice	Risk	Review question
Axis limits	Differences appear exaggerated or observations disappear	Does the scale support an honest comparison?
Transformation	Distances acquire a different interpretation	Is a log or other scale named and justified?
Aggregation	Within-group variation becomes invisible	What grain and summary rule produced each mark?
Binning	Distribution shape changes with break placement	Are conclusions stable across reasonable bins?
Overplotting	Dense observations conceal frequency	Would transparency, jitter, bins or facets help?
Free facet scales	Shapes are comparable but magnitudes are not	Does each panel need the same quantitative scale?

Never hide a consequential filtering, transformation or aggregation decision in plotting code.

Design for interpretation, access and reuse

Make meaning visible

- informative title and subtitle;
- axis labels with units;
- readable legends or direct labels;
- source and aggregation note; and
- annotation of the intended comparison.

Make the figure usable

- sufficient contrast and text size;
- colour-blind-safe palettes;
- meaning not carried by colour alone;
- appropriate dimensions and file type; and
- reproducible code and stable output path.

A self-contained figure states what is shown, how it was measured and what it cannot establish.

Ask complementary questions of prepared data

One outcome over entities and time

- distribution of the outcome;
- differences between entities;
- change over time; and
- unusual entity–period observations.

Outcome and a second variable

- distribution and trends of the predictor;
- outcome versus predictor;
- differences across relevant groups; and
- possible non-linearity or overplotting.

Each chart should state its observation grain: one mark may represent an entity–period, a group summary or an aggregated bin.

Turn an exploratory plot into a communication graphic

State one comparison → Generate and inspect alternatives → Check scale, grouping and access → Label, annotate and export

- Begin with a trend, distribution or relationship plot tied to one question.
- Compare faceting, grouping and scale choices against the intended question.
- Remove decoration that does not carry information.
- Add units, source, temporal alignment and a non-causal interpretation.
- Save the figure through code rather than manual export.

Key message: every visual choice shapes interpretation

Reproducible artifacts

- plotting script;
- exploratory figure set;
- one communication-ready figure;
- documented output paths; and
- source data and package environment.

Review evidence

- question matches the graphic;
- grain, units and scales are visible;
- grouping and aggregation are justified;
- design remains accessible; and
- interpretation respects limitations.

Visualization reveals and communicates patterns; the next session quantifies distributions and associations with descriptive statistics.

Show the evidence clearly—and make every consequential visual choice inspectable.

Supported by

South African government and the National Research Foundation

science, technology & innovation
Department: Science, Technology and Innovation
REPUBLIC OF SOUTH AFRICA

NRF National Research Foundation

German government and the German Academic Exchange Service

Supported by:
Federal Foreign Office

With funding from the:
Federal Ministry of Research, Technology and Space

Supported by:
Federal Ministry of Agriculture, Food and Regional Identity
by decision of the German Bundestag

Project Manager:
Federal Office for Agriculture and Food

DAAD
Deutscher Akademischer Austauschdienst
German Academic Exchange Service